
ARTICLE

Automated MRI-Based Brain Lesion Delineation Using Deep Convolutional Neural Models

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ABSTRACT: In medical image processing, Brain tumor segmentation plays an important role. Early detection of these tumors is highly required to provide Treatment for patients. The patient's life chances are improved by the early detection of it. The process of diagnosing brain tumour by physicians is normally carried out using a manual way of segmentation. It is a time-consuming and difficult one. To solve these problems, Enhanced Convolutional Neural Networks (ECNN) are proposed with loss function optimization by the BAT algorithm for an automatic segmentation method. The primary aim is to present optimization-based MRI image segmentation. Small kernels allow the design of a deep architecture. It has a positive consequence with respect to overfitting, provided the lesser weights are assigned to the network. Skull stripping and image enhancement algorithms are used for pre-processing. The experimental result shows better performance when compared with the existing methods. The compared parameters are precision, recall, and accuracy. In the future, different selection schemes can be adopted to improve the accuracy.

KEYWORDS: Brain tumor, segmentation, hybrid CNN, BAT algorithm, early detection, pre-processing, Standardizing Intensity Scales.

1 Introduction

Uncontrolled growth of cancer cells causes cancer in any part of the body. There are different types of tumors and their characteristics, as well as the treatments, differ in different ways [1]. Brain tumors are classified as primary and metastatic brain tumors. They started to grow in the brain and will stay in the brain. Later begins as a cancer in any part of the body and will spread to the brain.

Diagnosing the tumor in the primary stage can save lives significantly. Planning of treatment and intervention by the medical personnel depends on the Accurate segmentation of brain tumors. A qualified specialist requires a longer period of time to segment the tumors manually. So, it requires automatic segmentation and quantitative analysis of tumors [2].

High variation of Shape, structure, and location makes this process a challenging task. The arrangement of the surrounding normal tissues is changed by the tumor mass effect [3]. Intensity inhomogeneity [4] and different intensity ranges among the same sequences and acquisition scanners are produced in MRI images.



Diagnosing the tumors in the brain is usually done manually, way of segmentation by the experts. It is a time-consuming and difficult one. The major aim of this work is to introduce a new enhanced classification model to improve the tumor segmentation results. Enhanced Convolutional Neural Networks (ECNN) are introduced to resolve brain tumor segmentation. The BAT algorithm is used for automatic segmentation, which utilizes the loss function. Skull stripping and image enhancement [5] techniques are used to pre-process the MRI images. By using small kernels, deeper architectures are designed. Despite having a positive impact with respect to overfitting, provided the number of weights is less in a network [6].

2 Literature Review

Shen et al. [7] extended the traditional fuzzy [21]c-means (FCM) clustering algorithm to form a robust segmentation technique. Relative location and features of neighboring pixels define the neighborhood attraction, which is used to improve the segmentation performance dramatically. A neural network model is used to optimize the degree of attraction. But still, the accuracy of the classifier is not enhanced. The superiority of the proposed technique is compared to other FCM-based methods via real brain MR images with different noise levels. This is the currently deployed segmentation method for MRI images.

Kazerooni et al [8] modified the traditional GVF algorithm by applying the multi-scale idea to segment the brain tumors in MRI images. Scaled edge maps are used to evolve the active contour. Threshold-based edge detector is used to improve the performance of this method. Active contour is represented by selecting a B-spline snake. The B-spline snake is selected because of its ability to capture corners and its local control. 30% of accuracy improvement is achieved by using this method when compared to traditional GVF. Currently focusing on improving the robustness of the algorithm in high noise environments, but the detection rate has not improved.

Kalavathi and Ilakkiyamuthu [9] implemented a wavelet-based Fuzzy C-means (FCM) Clustering algorithm for segmentation. They have used Grey Level Co-occurrence Matrix (GLCM) feature extraction methods. Jaccard similarity and Dice coefficient values were used in evaluating the performance of the method that is currently proposed. Improved segmentation and classification of the tumors in the brain are attained using this method.

Par meet Knur and Harish Kundra [10] improved the intelligent water drops algorithm in order to detect the brain tumors. The IWD algorithm is used to detect brain tumors from the input MRI images. The improvement in the intelligent water drops algorithm is based on the SVM classifier. Cancer and non-cancer cells from the MRI images are classified by the SVM classifier. The output from IWD is fed to the SVM classifier. The performance of this system is detected in MATLAB. It provides a 20 % increase in the detection accuracy. The execution time is also reduced to 1.5 seconds.

Josephine [11] used the symmetry character of the brain image for segmentation of the Brain image. Automatic detection of tumor position and edge is the major goal of this work. Real images are used for experimentation. The experimental results show the flexibility and convenience of the proposed algorithm. It identifies the tumor region from T1, T2-weighted MRI brain images efficiently. The current classification methods are not applied to tumor types such as benign and malignant.

Yasmeen Khan and Anshul Bhatia [12] segmented the MRI image into the tumor region and the non-tumor region. Segmentation of the tumor area is done by utilizing Active Contour calculation. Early preparation is not required in this method. Median filters are used to remove the noise from the image. Different parameters are used for assessment and comparison with existing methods. The results have proven its efficiency.

Chen et al [13] combined fuzzy clustering and MRF to segment the MRI image. Fuzzy clustering is done on the original image in order to utilize the gray level information and to eliminate noise

disturbance. It is also based on the coarse-scale image of multi-scale decomposition. MRF is used to improve the integrity of the segmented image and to reduce the noise effect. Maximum a Posteriori Markov Random Field (MAP-MRF) is used to integrate the spatial constraints and the gray level information.

3 Proposed Methodology

The proposed architecture of the brain tumor segmentation is discussed in this section. There are two phases of operation: i. pre-processing, ii. Segmentation. Skull stripping and image enhancement methods are used in pre-processing. A Hybrid Convolutional Neural Network (HCNN) is used for segmentation. Loss function is optimized using the BAT algorithm. Figure 1 shows the proposed architecture.

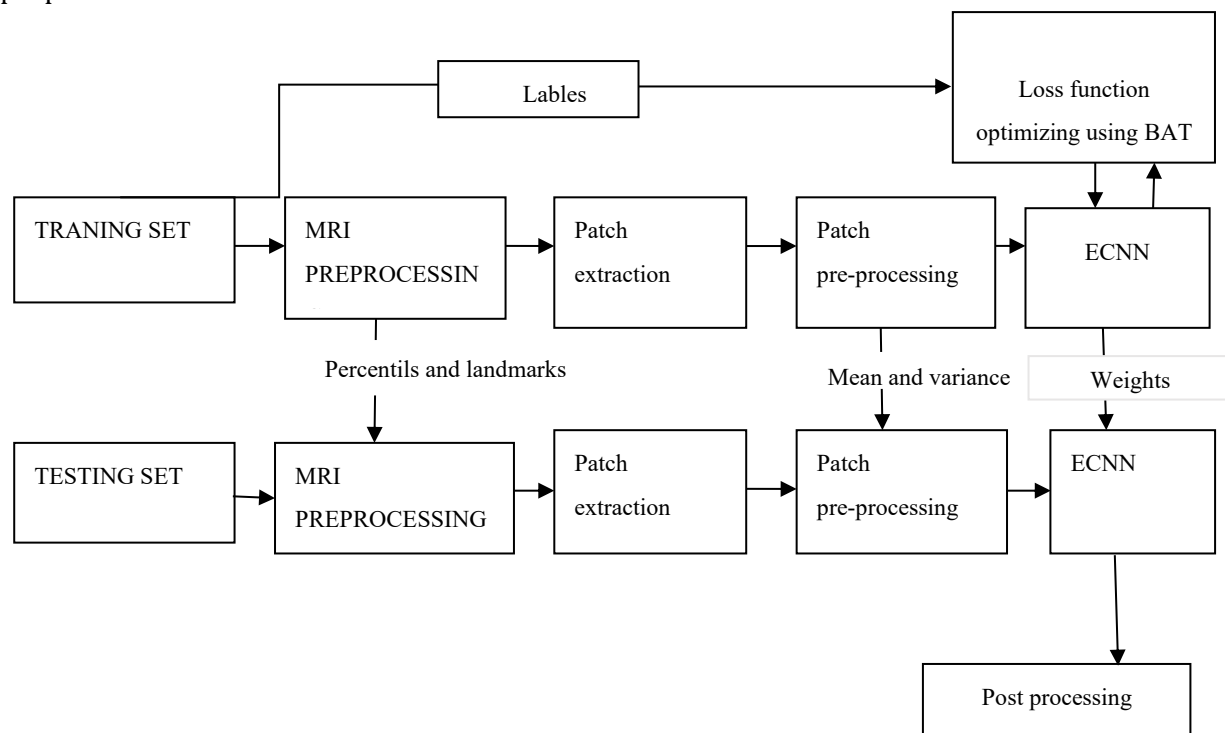


Figure 1: Overall Architecture of the Proposed Model

Various techniques for Pre-processing are available for dealing with the MRI images of the brain. Here, the MRI images get converted into a grey type of image, which is again smoothed by adjusting the contrast.

3.1 Skull Stripping

This is one of the popular pre-processing techniques that produces improved results. The surroundings of the brain are termed the skull. The Skull stripping is the process of eradicating the tissues that are not cerebral. It is difficult to distinguish the non-cerebral and the intracranial tissues because of their homogeneity in intensity. This is the major problem of skull stripping. Seed point selection is affected by this. To identify the range in which the grey value lies in a particular portion of the skull, observations are mandatory. Binary images are constructed using contrast adjustment. Crop locations are found in the binary images. The images with their contrasts adjusted are then cropped to find the part where the tumor is located in the brain. Cropped contrast adjusted image is converted into binary using a low threshold value and also converted into binary image.

Morphological operation 'thicken' is applied to this binary image. Region-based binary mask extraction is used to extract the brain region [14].

3.2 Region-Based Binary Mask Extraction

Properties of each block that satisfy some criteria are used to perform the Region-based extraction. Max-min difference and mean values of the blocks are the most widely used criteria. The brain mask is applied to the original MRI data. This produces the brain image with the cortex stripped.

3.3 Image Enhancement

The resultant images [22] are made suitable for display or further analysis by using Image enhancement techniques. It is termed the process in which adjustments are made for a digital image. After the process of skull stripping, the cortex of the brain can now be viewed as a unique dark ring that surrounds the brain in an MRI image. Image-enhanced methods are used to remove the distinct dark ring surrounding the brain tissues. For example, key features are identified by using the noise removal process. Morphological operation 'thicken' is applied to the binary image. Mean intensity value and standard deviation of all training patches extracted for each sequence after normalization. Patches on each sequence are normalized to have zero mean and unit variance. An Enhanced Convolutional Neural Network (ECNN) is used to segment the brain image after pre-processing.

3.4 Construction of the Central-Point-Enhanced Layer

Feather maps are classified by using the dominant point due to the CNN's invariant characteristics. The results from the segmentation process of CNN might not match the brain tumor category if the feather map contains the pixels that belong to other brain parts, which are non-brain categories [15].

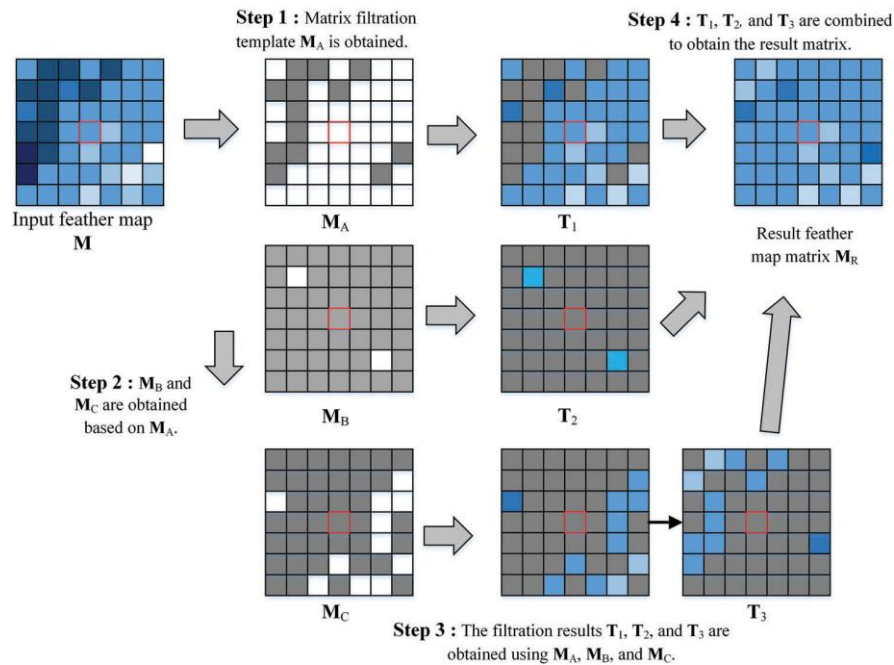


Figure 2: Processing of the central-point-enhanced algorithm

A new layer is constructed for the CNN to solve this problem. Consider the following aspects:

(1) The layers that are designed freshly have to filter out the pixels that are different from those of the tumor, so that the dominant point exists for the prevention of the decision being affected. This layer must have the capacity to preserve the exact texture or the characters in the feather map.

(2) The calculation on the matrix is used for the integration of the new design layer along the structure of the CNN. It is basically because of the fact that the GPU's acceleration, which is involved in the process of calculation in a CNN which works the Matrix calculation instead of using the process of conditional branching. Figure 2 depicts the algorithm that is a central point enhanced.

In Figure 2, the layer with an enhanced central point is generated for each input feather map. The (CE-Alg) is defined as

Input: The feather map matrix denoted by m and the difference threshold represented by α

Output: The m_R , which is the result of the feather map matrix.

Step 1: Get the template of the matrix filtration represented as m_A .

The difference between an element and the central point is computed using (1)

$$diff(x) = (1 - e^{-5(x-x_{centre})^2}) \quad (1)$$

where x defines an element in matrix m , the central point of matrix m is given by x_{centre} . The value of $diff$ approaches 0 as the value of x approaches the central point's value. The value gets increased with the increase in the difference in-between x to the central point. The Matrix of m_A is gained using (2),

$$m_A = diff(m) < \alpha \quad (2)$$

After the calculation of the $diff$ of all the elements in m , they are compared with the logical '<' expression. In the resultant matrix, 0 denotes 'false' and 1 denotes 'true'. The m_A matrix is constructed using this way. The value '0' corresponds to the high difference from the central point, and the value '1' denotes the small difference between the central point. In Figure 2, m_A , '0' and '1' represent the grey and white colours respectively.

Step 2: Based on m_A , m_B , and m_C are computed. Diagonal matrix J is constructed with width and height equal to the width and height of m :

$$J = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ \cdot & \cdot & \cdot & \cdot \\ 1 & 0 & 0 & 0 \end{bmatrix} \quad (3)$$

m_B is obtained as follows,

$$m_b = ((m_a - 1)) + (j \cdot m_a \cdot j) - 1 + 2) == 0 \quad (4)$$

Logical equals comparison is denoted by ' $==$ ' and the dot product of a matrix is denoted by ' $\cdot$ '. In m_B , logical 'true' corresponds to '1' and logical 'false' corresponds to '0'. Once the rotation of the elements in m_A is done surrounding the central point in m_B , the positions that remain '0's are marked with '1', and the rest of the elements are marked as '0'. The matrix m_C is constructed based on m_A and m_B as,

$$m_c = -((j \cdot m_a \cdot j) - 1) - m_B \quad (5)$$

After rotating m_A around the central point in m_C , the positions with the value equals '1' corresponds to positions '0' in m_B and marked with '1' all the other elements are marked '0'.

Step 3: m_A , m_B , and m_C are filtered to obtain results t_1 , t_2 , and t_3 , mis-filtered through the m_A . The values that are retained correspond to the elements that are similar to the central point in a feather map.

$$t_1 = m \times m_a \quad (6)$$

Elements in the matrix are then multiplied separately by the position with the use of '×'. Matrix t_2 is obtained by assigning the value of the central point to obtain the '1' in m_B as follows

$$t_2 = m_b \times x_{centre} \quad (7)$$

The Filtration process of m is executed through m_c . Filtered results are then rotated surrounding the central point. The t_3 type of Matrix is attained using (8),

$$t_3 = j.(m \times m_c) \quad (8)$$

Step 4: Result matrix is obtained by combing t_1 , t_2 , and t_3 . The resulting feather map matrix m_r is calculated as follows:

$$m_r = t_1 + t_2 + t_3 \quad (9)$$

Elements having huge differences as the number of central points in a feather map, are filtered using this preceding algorithm. Corresponding positions are filled by pixels that are in the positions of symmetry with that of the central point. If one happens to filter the elements that are present in the symmetric positions, then the CPV "Central Point Value" is used for filling the element. Processing for the CE-L is shown as follows:

Processing for CE-L
 Input: Feather map Set $S=\{M_1, \dots, M_n\}$, distance threshold α
 Output: Feather map Set $R=\{R_1, \dots, R_n\}$
 Begin
 Step 1: $R = \varphi$
 Step 2: for i in 1:N
 $R_i = CE - alg(M_i, \alpha)$
 $R_i = R \cup R^i$
 End for
 Step 3: return R
 End

CE-Alg processes every input feather map in I . All feather map sets are formed by input and output CELs. CE-Alg involves the matrix calculations. This layer can be integrated effectively with a CNN structure by using this characteristic.

3.5 Structure of the CE-CNN

Figure 3 depicts the primary blocks, the input, process, and the output in a CEL. The brain tumor image includes the set of images n whose width is w and height is h . This set of brain images is normalized onto the interval $[0, 1]$. Consider the pixel p with location (x,y) . The central point of a square neighbourhood region is denoted as p . It is constructed with one pixel. The square region's width and height are $L = 2 \times 1 + 1$. The image patch is cut from the image based on a neighbourhood square region, and it is used to construct the feather map set $S = \{M_1, M_2, \dots, M_n\}$. Each set of similar images corresponds to one feather map M_i . The input of the CE-CNN is given by S , and it is processed as,

(1) Central-point-enhanced layer: for all the feather maps inside S , an improved feather map set $R = \{R_1, R_2, \dots, R_n\}$ is constructed using CE-Alg.

(2) Convolutional layer: W no of feature maps are gained through W kernels with sizes of $K \times K$ by convolution calculation.

(3) Max-pooling layer: a maximum W input feature maps with each of $n \times n$ block is conducted. Critical characteristics are obtained by reducing the size of the feather map.

(4) Flatten layer: The previous layer's output feature maps are reshaped. The neurons obtained as output are then changed to gain a one-dimensional vector.

(5) Fully connected three-layer MLP: This is a multi-layered network that is neural and has all the input, hidden, and output layers. The first layer is connected directly to the layer that is flat. The count of the neurons present in the hidden layer is double that of the one which is in the output layer.

The transmission function used here is ReLU. A more complicated and non-linear pattern in samples is expressed by the hidden layer. The MLP's ability to segment input samples is highly influenced by the number of neurons in the hidden layer.

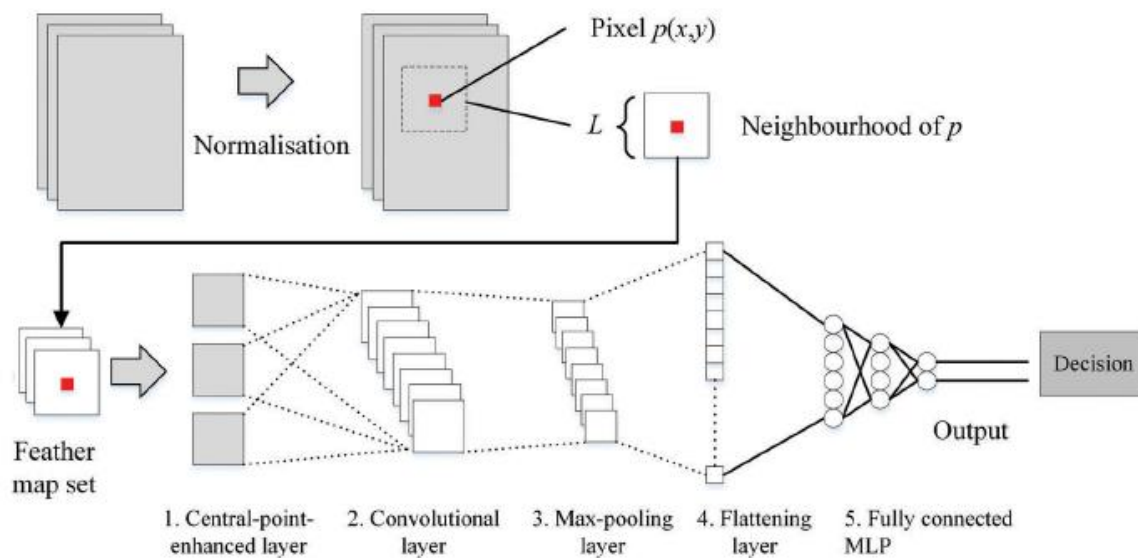


Figure 3: Structure, input, and output of the CE-CNN

A minimum no of nodes present in the hidden layer will lead to the non-adequate expression of the neural network, which affects the accuracy of the segmentation.

Large numbers of nodes will lead to overfitting to training data. That reduces the network's generalization ability. So twice the number of neurons in the output layer is set as the default number of nodes in the hidden layer. MLP was capable of expressing the pattern inside the brain tumor test samples and refrained from overfitting. The count of the criteria will equal the count of the output layers. The transmission function used here is the Softmax. The ability in terms of the pixel in a CNN is improved by CE-CNN [16]. In the second section of the CE-CNN, the tiniest kernels (3×3 in size) are selected in the convolutional layer. The third section of the CE-CNN deals with the max-pooling by the selection of the smallest kernel. These two layers are then used to obtain the spatial data. Reducing the dimensions of the feather map that is taken as input will help the consequent CE-CNN for predicting the sample more efficiently.

Single-group convolutional/max-pooling layers were required to achieve this objective when the input feature map was small. Input feather map set $S = \{M_1, M_2, \dots, M_n\}$ is processed through these five parts to obtain the corresponding brain tumor categories for the pixel p . During the training stage for the CE-CNN, each training sample pixel is transformed into a training feather map set. Each

pixel of the image is transformed into the feather map set at the image segmentation stage. The category is obtained by the learned segmentation model. The pixel's category is written into the corresponding pixel's position in the resulting image. The prediction error is computed using a loss function like categorical cross-entropy. Once the forward pass is complete, the backpropagation begins. This is used to update the weights and biases for error and loss reduction. The BAT optimization algorithm is used to optimize the loss function.

3.6 Novel Bat Optimization Algorithm (NBOA) for Loss Function

The ECNN output represents the estimated probability of a pixel belonging to a skin tumor. ECNN performs a pixel-wise segmentation. The loss function used here is cross-entropy.

A new Novel Bat Optimization Algorithm (NBOA) is proposed to reduce the errors in the brain tumor segmentation. The difference between a non-optimal (obstacle) and an optimal error (prey) is decided based on the Echolocation mechanism. It allows them to hunt even in total darkness [17-18]. The ability of echolocation of bats in NBOA methods is used to reduce the error $SU = SU = \{f_1, f_2, \dots, f_m\}$ from the brain tumor image dataset. In NBOA, the echolocation distinctiveness is idealized by the following rules. These rules are formed by the reduction of errors by bats [19]:

Segmentation accuracy is sensed by echolocation by bats. Bats also know the difference between higher brain tumor segmentation accuracy (prey/food) and surrounding barriers.

The velocity, feature position, frequency, loudness, and wavelength of a bat are given by vel_i , fp_i , $freq_{min}$, A_0 , and γ . Bats fly randomly to search for improved tumor segmentation accuracy. Depending on the closeness of their Segmentation accuracy, wavelength, and emitted pulses are fine-tuned.

Loudness can fluctuate in many ways. It ranges from a large value A_0 to the smallest value A_{min} .

For every bat b_i , starting feature position fp_i , velocity vel_i , and frequency $freq_i$ are initialized. For each iteration t , the movement of the virtual bats is specified. This is done by updating their velocity and position. They are given by equations (10), (11), and (12) as follows: T defines the maximum number of iterations,

$$freq_i = freq_{min} + (freq_{max} - freq_{min})\beta \quad (10)$$

$$vel_i^t = vel_i^{t-1} + (fp_i^t - fp_*)freq_i \quad (11)$$

$$fp_i^t = fp_i^{t-1} + v_i^t \quad (12)$$

Random vector $\beta \in [0, 1]$ is drawn from a uniform distribution. Frequency of each bat is defined by $freq_i$. Current global optimal error is given by **fp**. It is located after comparing all the solutions among all n errors at each iteration ' t '. A random number is created after the error position updating of bats. A fuzzy membership function is used for the same. A new error position will be created around the current best selected error if the random number is larger than the pulse emission rate r_i . It is given by (13),

$$fp_{new} = fp_{old} + \epsilon A^t \quad (13)$$

where $\epsilon \in [-1, 1]$ is a random number. The average loudness of all the bats at the current iteration is given by A^t . The loudness A_i and $f(fp_i) < f(fp)$. A_i , the pulse emission rate r_i will be updated, and a solution will be accepted if a random number is less than loudness A_i and $f(fp_i) < f(fp)$. A_i and r_i are updated by (14-15),

$$A_i^{t+1} = \alpha A_i^t \quad (14)$$

$$r_i^{t+1} = r_i^0 [1 - \exp(-\gamma t)] \quad (15)$$

where α, γ are constants. The algorithm iterates until the termination criteria are met. In the proposed NBOA algorithm, the parameters of the bats, such as $\beta, \epsilon, A_i, \gamma$, and r_i , are tuned via the use of a Gaussian fuzzy membership function (GFMF). The Gaussian membership function is usually represented as $\text{Gaussian}(x; \mu, \sigma)$, where μ, σ represent the mean and standard deviation,

$$F_A(x, \mu, \sigma, m) = \exp \left[-\frac{1}{2} \left| \frac{x - \mu}{\sigma} \right|^m \right] \quad (16)$$

where m belongs to the Fuzzification parameter ($m=2$).

Algorithm for Novel Bat Optimization Algorithm (NBOA)

1. Bat population is initialized fp_i ($i=1,2,\dots,n$) and vel_i
2. The pulse frequency $freq_i$ at fp is defined
3. The pulse rates r_i along with the loudness A_i are defined
4. While ($t < \text{Max number of iterations}$)
 - 4.1. Generation of fresh solutions by frequency adjustments and update velocities /locations and solutions [equations (10) to (12)]
 - 4.2. if ($\text{rand} > r_i$)
 - 4.2.1. The optimal error column is selected
 - 4.2.2. Local error is generated surrounding the best errors that are selected
 - 4.3. end if
 - 4.4. Generate a new solution by flying randomly
 - 4.4.1. if ($\text{rand} < A_i$ & $f(fp_i) < f(fp^*)$) //compute A_i via fuzzy membership function.
 - (i) Accept the new solutions
 - (ii) Increase r_i and reduce A_i
 - 4.4.2. end if
 - 4.5. Rank the bats and find the current best x^*
5. end while
6. Postprocess results and visualization

It can be efficiently integrated into the backpropagation during network training.

4 Results and Discussion

This section discusses the experimental results of the proposed model. The model is implemented using Java. The existing Convolutional Neural Network (CNN) algorithm [20] and proposed Enhanced Convolutional Neural Network (ECNN) are compared in terms of precision, recall and accuracy for the dataset of the Brain Tumor Segmentation Challenge 2015 database (BRATS 2015), obtaining simultaneously the first position for the complete, core, and enhancing regions in Dice Similarity Coefficient metric for the Challenge data set.

4.1 Performance Metrics

4.1.1 Precision

Precision refers to the percentage of the relevant results, which is defined as,

$$\text{Precision} = \frac{\text{Truepositive}}{\text{truepositive}+\text{falsepositive}} \quad (17)$$

4.1.2 Recall

Recall refers to the percentage of total relevant results correctly classified by the proposed algorithm, which are defined as,

$$\text{Recall} = \frac{\text{True positive}}{\text{true positive}+\text{False Negative}} \quad (18)$$

4.1.3 Accuracy

Accuracy is one metric for evaluating classification models. Informally, accuracy is the fraction of predictions this model got right. Formally, accuracy has the following definition:

$$\text{Accuracy} = \frac{\text{Truepositive}+\text{TrueNegative}}{\text{Total}} \quad (19)$$

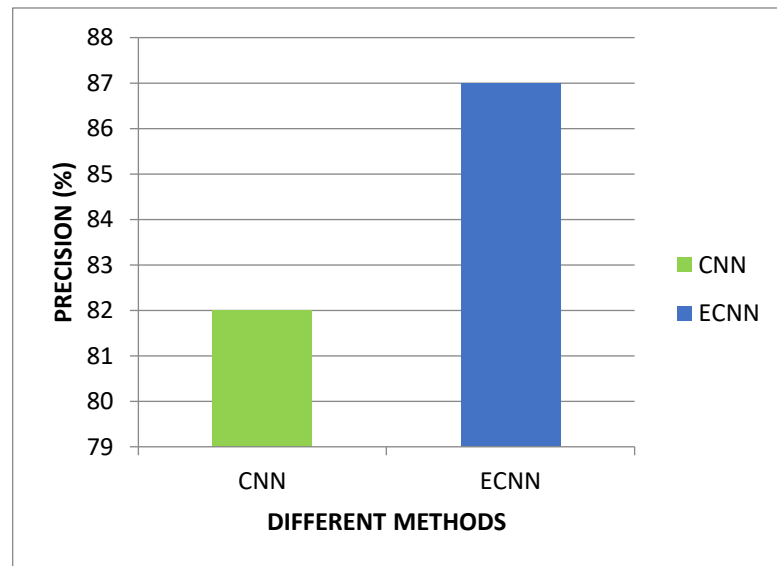


Figure 4: Precision results of different methods

Figure 4 shows the performance comparison results of the proposed ECNN method and the existing CNN methods in terms of Precision. From the results, it is concluded that the proposed ECNN model produces higher precision results of 87% whereas the existing CNN method produces only 82%.

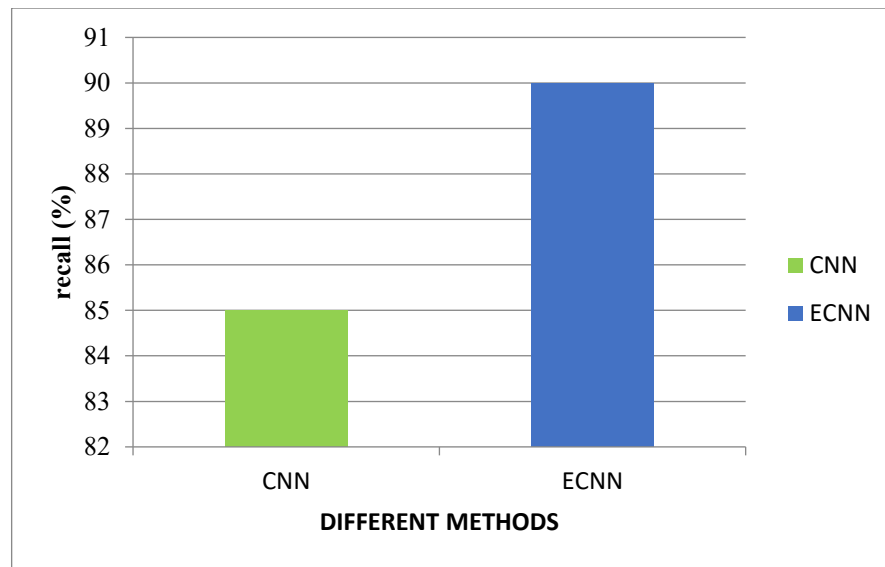


Figure 5: Recall results of different methods

Figure 5 shows the performance comparison results of the proposed ECNN method and the existing CNN methods in terms of Recall. From the results, it is concluded that the proposed ECNN model produces higher recall results of 90% whereas the existing CNN method produces only 85%.

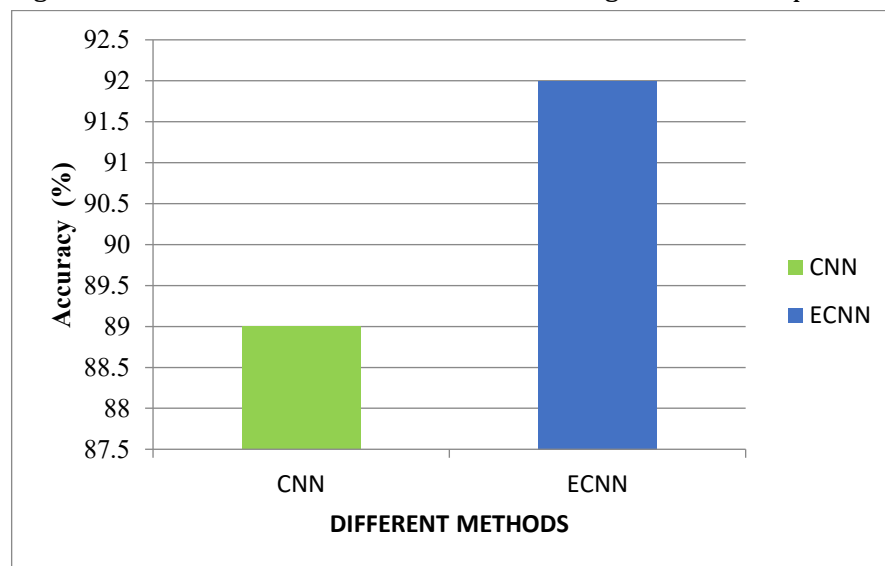


Figure 6: Accuracy results of different methods

Figure 6 shows the performance comparison results of the proposed ECNN method and the existing CNN methods in terms of accuracy. The results indicate that the proposed ECNN model. From the results, it is concluded that the proposed ECNN model produces higher accuracy results of 92% whereas the existing CNN method produces only 89%.

5 Conclusion and Future Work

Enhanced Convolutional Neural Networks (ECNN) with loss function optimization by the BAT algorithm are used in order to perform segmentation of an MRI image in an automated manner. Tiny

kernels are used to achieve a deep architecture. It has a positive effect against overfitting, given the smaller number of weights in the network. Skull stripping and image enhancement algorithms are used for pre-processing. The experimental results show better performance when compared with the existing methods. The compared parameters are precision, recall, and accuracy. In the future, different selection schemes can be adopted to improve the accuracy.

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